Abstract

This study, *FEMCARE: A Research Based Approach to Improving* ***Women's Health and Hygiene***, examines menstrual health challenges among women in India through a data driven research framework. Menstrual health continues to remain one of the most overlooked aspects of women's well being, particularly in relation to hygiene awareness, access to healthcare, and the availability of safe menstrual products. Many women continue to face preventable health complications due to social stigma, limited education, and inadequate healthcare support.

The research is based on a combined dataset of **10,087 respondents**, created using both a l**arge public dataset obtained from online repositories and primary survey data collected through a structured Google Forms questionnaire**. The dataset contains **42 variables** covering menstrual cycle patterns, hygiene practices, emotional health, product usage, medical diagnoses, and socioeconomic factors.

The analytical process was carried out in Python using libraries such as Pandas, NumPy, Matplotlib, and Seaborn. The study involved data cleaning, exploratory data analysis, correlation analysis, and statistical hypothesis testing using T tests, Chi square tests, Pearson correlation, and one way ANOVA.

The findings show that nearly **30% of respondents experience irregular menstrual cycles, with chronic fatigue and hormonal imbalance emerging as major contributing factors.** The prevalence of PCOS was observed at 18%, which is higher than many commonly reported national estimates. In addition, **54% of respondents had not consulted a doctor in the previous year, highlighting a significant gap in healthcare accessibility.**

Statistical analysis identified a positive relationship between anxiety levels and menstrual pain severity. Poor hygiene practices were also strongly associated with a higher history of infections. **The type of menstrual product used was found to significantly affect infection rates, with cloth users reporting the highest prevalence of infections.** The study further revealed that **women living in urban areas demonstrated higher levels of hygiene awareness and medical consultation rates compared to women in rural regions**, indicating clear geographical disparities in awareness and healthcare access.

All five research hypotheses were statistically supported at the **0.05 significance level**. Based on these findings, the study proposes practical recommendations focused on improving menstrual hygiene education, strengthening rural healthcare access, increasing awareness about PCOS, integrating mental health support, and promoting the distribution of safe menstrual products. The research demonstrates how data science techniques applied to real world health datasets can generate meaningful insights that may support better healthcare policies and improve women's health outcomes.

Introduction

1.1 Background and Context

Menstruation is a natural biological process experienced by nearly **1.8 billion women and girls** worldwide every month. Despite being a normal part of life, menstrual health continues to be affected by stigma, misinformation, and limited healthcare attention, especially in developing countries such as India. Inadequate menstrual hygiene management, poor access to safe menstrual products, lack of sanitation facilities, and low awareness about conditions like **Polycystic Ovary Syndrome (PCOS)** and endometriosis contribute to avoidable health complications and reduced quality of life.

In India, reports suggest that only **36% of women use sanitary napkins** during menstruation according to NFHS-5 (2019–21). A large number of women still depend on reused cloth, which increases the risk of infections and other reproductive health issues. Menstrual pain, commonly known as dysmenorrhea, also affects the daily lives of millions of women by causing discomfort, absenteeism from school or work, and emotional stress. Although menstrual health has a direct impact on physical and mental well being, it remains insufficiently represented in national health research and policy discussions.

1.2 Research Problem

One of the major challenges in menstrual health research is the lack of large scale, data driven studies that examine multiple dimensions of women's health together. Many existing studies are restricted by small sample sizes, limited regional coverage, or focus on only one aspect of menstrual health. As a result, healthcare providers, educators, and policymakers often lack reliable evidence for designing effective interventions.

There is a growing need for integrated research that studies menstrual hygiene practices, emotional health, clinical conditions, product usage, awareness levels, and healthcare seeking behavior together. A comprehensive statistical approach can help identify hidden patterns and relationships that may otherwise remain unnoticed.

1.3 Objectives of the Study

The main objectives of this study are:

1. To analyze menstrual health patterns such as cycle regularity, pain severity, and flow intensity among women from different demographic groups.
2. To examine the relationship between hygiene practices and health outcomes, particularly infection history and menstrual product usage.

3. To investigate the association between psychological factors such as anxiety, depression, PMS severity, and menstrual pain intensity.
4. To assess awareness regarding ovulation, PCOS, and healthcare consultation behavior among urban and rural respondents.
5. To statistically test the proposed research hypotheses using appropriate inferential techniques.
6. To generate practical and data supported recommendations for improving women's menstrual health and hygiene outcomes.

1.4 Main Research Question

How can data driven analysis help identify the causes of menstrual health issues and support women in managing their menstrual cycle more effectively?

1.5 Sub Questions

- What factors influence irregular menstrual cycles?
- How do hygiene practices affect health outcomes?
- Is there a relationship between lifestyle habits and menstrual disorders?
- Can predictive insights support the early identification of conditions such as **PCOS**?

1.6 Scope and Delimitations

This study is based on a combined dataset of **10,087 respondents**, collected from a public online repository and a primary Google Forms survey. The research follows a **quantitative and cross sectional** approach. The study focuses on statistical analysis and does not involve clinical trials or long term medical tracking.

The findings are primarily descriptive and correlational in nature, meaning that relationships between variables can be identified, but direct causation cannot be confirmed. The dataset includes respondents from multiple Indian states and represents both **urban and rural populations**.

1.7 Significance of the Study

This study contributes to the growing field of **data driven women's health research** by demonstrating how large datasets can be used to generate meaningful insights into menstrual health challenges. The findings may help NGOs, healthcare providers, educational institutions, and public health departments design evidence based awareness programs and healthcare interventions.

From an academic perspective, the project also demonstrates a complete **data science workflow**, including data collection, cleaning, exploratory analysis, statistical testing, and interpretation of results within an important public health domain.

Review of Literature

Existing research on menstrual health has expanded significantly over the last decade, especially in low and middle income countries where menstrual hygiene, awareness, and healthcare access remain major public health concerns. The following studies and reports provide the conceptual and empirical foundation for the present research.

3.1 Menstrual Hygiene Management in India

The **National Family Health Survey (NFHS-5, 2019–21)** reported that nearly **64% of Indian women aged 15–24** use hygienic menstrual protection methods. However, the survey also highlighted a strong urban rural gap, with usage rates of **78% in urban areas** compared to **57% in rural areas**. Limited affordability, restricted access to products, and cultural stigma were identified as major barriers to proper menstrual hygiene management. These findings provide important background for the hygiene and product accessibility analysis conducted in this study.

3.2 PCOS Prevalence and Awareness

According to **Ndefo et al. (2013)**, **PCOS** affects approximately **6–10% of women of reproductive age** worldwide, making it one of the most common endocrine disorders among women. Indian research, however, has reported even higher prevalence rates ranging from **9.13% to 22.5%** depending on diagnostic criteria and population groups studied (**Joshi et al., 2014**).

Several studies also emphasize the issue of delayed diagnosis, with many women experiencing a gap of **2–3 years** between the onset of symptoms and medical confirmation. Low awareness levels and inadequate healthcare consultation were identified as key reasons behind this delay. These findings support the awareness and healthcare accessibility analysis explored in the present study.

3.3 Psychological Dimensions of Menstrual Health

Research conducted by **Iacovides et al. (2015)** found that dysmenorrhea, or painful menstruation, is strongly associated with anxiety, depression, and reduced quality of life. Their review of 20 separate studies showed that women experiencing severe menstrual pain consistently reported higher anxiety levels compared to control groups.

The study proposed a psychosomatic relationship between emotional stress and pain severity, suggesting that mental health factors may significantly influence menstrual experiences. This literature supports the investigation of the relationship between anxiety levels and menstrual pain severity in the current research.

3.4 Menstrual Product Use and Infection Risk

A systematic review by **Sumpter and Torondel (2013)** examined menstrual hygiene practices across 14 studies and found clear evidence linking improperly cleaned reusable cloth materials with higher rates of reproductive tract infections (RTIs). Research conducted in rural India and Sub

Saharan Africa reported infection rates that were **2–3 times higher among cloth users** compared to women using disposable sanitary pads.

These findings provide a strong foundation for examining the relationship between menstrual product usage, hygiene practices, and infection history within the present dataset.

3.5 Data Science Applications in Women's Health

The use of **data science and machine learning in menstrual health research** has emerged as a rapidly developing field. **Symul et al. (2019)** demonstrated that predictive models trained on self reported menstrual cycle data could estimate cycle length and ovulation timing with reasonable accuracy.

Recent studies using large scale data from period tracking applications such as Clue and Flo have also identified broader population level trends related to cycle variability, PMS severity, and reproductive health conditions. Inspired by these approaches, the present study applies statistical and data driven methods to analyze menstrual health patterns within an Indian demographic context.

3.6 Urban Rural Disparities in Menstrual Health

Kaur et al. (2018) studied menstrual health awareness among adolescent girls in rural Punjab and found that nearly **72% of participants had no knowledge of menstruation before menarche**. The research also reported significantly poorer hygiene practices in rural communities compared to urban populations.

The study recommended community based education programs as an essential intervention for improving menstrual health awareness and hygiene practices. These findings align closely with the urban rural disparities identified in the current research dataset.

3.7 Research Gap Addressed by This Study

Although previous studies have explored individual aspects of menstrual health such as hygiene practices, PCOS prevalence, menstrual pain, or product usage, very few studies have examined all these dimensions together within a single analytical framework. Additionally, many Indian studies are based on relatively small sample sizes, often involving fewer than 500 participants.

The present study addresses this gap by combining a large-scale secondary dataset containing **9,999 records** with primary survey data collected from **88 respondents**. Using a comprehensive statistical analysis pipeline across **42 variables**, this research provides a broader and more integrated understanding of menstrual health patterns, awareness levels, healthcare access, and hygiene related outcomes among women in India.

Methodology

4.1 Research Design

This study follows a **quantitative cross sectional research design**. Data was collected from two different sources and analyzed using descriptive as well as inferential statistical techniques. The research combines both exploratory and confirmatory approaches, aiming to identify menstrual health patterns while also testing predefined hypotheses using statistical evidence.

4.2 Data Sources

4.2.1 Online and Public Dataset

A structured public dataset was obtained from **Kaggle** and other open data repositories. The dataset contains **9,999 records** related to women's menstrual health, including variables associated with cycle patterns, pain levels, hygiene practices, product usage, emotional health indicators, and clinical diagnoses.

The dataset was selected because of its large sample size, diversity of variables, and strong alignment with the objectives of the study.

4.2.2 Primary Survey Dataset

Primary data was collected through a structured **Google Forms questionnaire** distributed digitally through college networks and social media platforms. The survey consisted of **42 questions** designed to match the variables available in the public dataset, allowing both datasets to be merged efficiently.

4.2.3 Combined Master Dataset

After standardizing column names and variable structures, both datasets were merged into a master dataset named **Combined.csv**. The final dataset contains **10,087 records across 42 variables**, which served as the primary source for all statistical analysis and visualizations performed in this study.

4.3 Tools and Technologies

The following tools and technologies were used throughout the project:

- **Python 3.12** for programming and data analysis
- **Pandas** for data cleaning, transformation, and manipulation
- **NumPy** for numerical operations
- **Matplotlib** and **Seaborn** for data visualization
- **SciPy (stats module)** for statistical hypothesis testing
- **Jupyter Notebook** for interactive analysis and experimentation

4.4 Data Cleaning Process

The combined dataset underwent multiple preprocessing and cleaning steps before analysis:

1. **Duplicate Detection and Removal**
   Duplicate records were checked across the dataset. No duplicate entries were found, and all **10,087 records** were retained.
2. **Data Type Correction**
   Three variables — *Average_Cycle_Length_Days*, *Monthly_Spending_INR*, and *Late_Cycle_Frequency_Per_Year* — were converted from object format to numeric format to support statistical analysis.
3. **Text Value Mapping**
   Non numeric survey responses such as "Twice" and "Nil" in the *Late_Cycle_Frequency_Per_Year* column were converted into corresponding numeric values.
4. **Categorical Standardization**
   Inconsistent category labels were standardized. For example, "Post-Graduate" was converted to "Postgraduate," and "Menstrual Cup" was shortened to "Cup." Income related values were grouped into five standardized income categories.
5. **Missing Value Handling**
   Columns with high missing values such as *Product_Access_Difficulty* (**53.6% missing**) and *Medication_Usage* (**29.2% missing**) were filled using "Not Specified." Numeric columns with less than 1% missing values were imputed using the median, while categorical variables with low missing rates were filled using the mode.
6. **Outlier Analysis**
   Box plot analysis was performed on 10 major numeric variables to detect extreme values. All outliers were retained because they represented valid biological or behavioral variations rather than data entry errors.
7. **Final Dataset Preparation**
   The cleaned dataset was saved as **Femcare_cleaned.csv**, containing **10,087 rows and 42 columns** with no remaining null values.

4.5 Analytical Framework

The analytical process was divided into four major stages:

Stage 1: Descriptive Analysis

This stage focused on dataset overview, summary statistics, and distribution analysis of both numeric and categorical variables.

Stage 2: Exploratory Data Analysis (EDA)

A series of **12 targeted visual analyses** were performed to examine patterns related to menstrual cycle regularity, pain severity, hygiene practices, product usage, PCOS prevalence, anxiety levels, emotional health, infection history, and productivity impact.

Stage 3: Correlation Analysis

A full **Pearson correlation heatmap** was generated using more than 25 encoded variables. Additional scatter plots, box plots, and grouped bar charts were created to analyze relationships between key hypothesis related variables.

Stage 4: Hypothesis Testing

Formal statistical testing was conducted for all five research hypotheses using:

- **T tests** for group mean comparisons
- **Chi square tests** for categorical associations
- **Pearson correlation** for measuring linear relationships
- **One way ANOVA** for comparing multiple groups

All hypotheses were tested at a **significance level of α = 0.05**.

4.6 Ethical Considerations

Participation in the Google Forms survey was completely voluntary and anonymous. No personally identifiable information was collected from respondents. Participants were informed that the survey was intended solely for academic research purposes.

The public dataset used in this study was openly available for academic and research use. No medical procedures, clinical testing, or direct interventions were conducted during the research process.

Results and Discussion

This section presents the major findings obtained from descriptive statistics, exploratory data analysis, correlation analysis, and statistical hypothesis testing. The results are supported using visualizations, statistical evidence, and research based interpretations to better understand the patterns and factors influencing women's menstrual health.

The cleaned master dataset consists of **10,087 respondent records** distributed across **42 variables** related to menstrual health, hygiene practices, emotional well being, clinical conditions, and healthcare behavior. The dataset includes participants between the ages of **13 and 55 years**, with the majority belonging to the **19–35 age group**.

In terms of geographic distribution, approximately **58% of respondents were from urban areas**, while **42% represented rural populations**. The dataset was created by combining **9,999 records from public datasets** with **88 valid responses collected through the primary survey**, providing both large scale statistical coverage and additional contextual insights.

The diversity of variables and demographic representation enabled a broad analysis of menstrual health patterns across different population groups.

5.2 Descriptive Statistics

The descriptive analysis provides an overview of the major numerical variables within the dataset.

- **Mean Age:** approximately 28 years
- **Median Age:** 27 years
- **Average Menstrual Cycle Length:** approximately 28 days, which falls within the medically accepted normal range
- **Mean Pain Level:** approximately 5.0 out of 10, indicating moderate menstrual pain on average
- **Mean PMS Severity:** approximately 5.0 out of 10
- **Mean Anxiety Level:** approximately 5.0 out of 10
- **Mean Hygiene Knowledge Score:** approximately 5.5 out of 10
- **Average Missed Work or School Days:** nearly 2 days per menstrual cycle

These findings suggest that while the average menstrual cycle length remains within standard medical ranges, many respondents continue to experience moderate levels of menstrual pain, PMS symptoms, anxiety, and productivity disruption. The hygiene knowledge score also indicates that awareness levels remain moderate rather than highly informed, highlighting the continued need for menstrual health education and awareness initiatives.

Figure 6.1 — Respondent Age Group Distribution

Figure 6.1a — Age Group Distribution

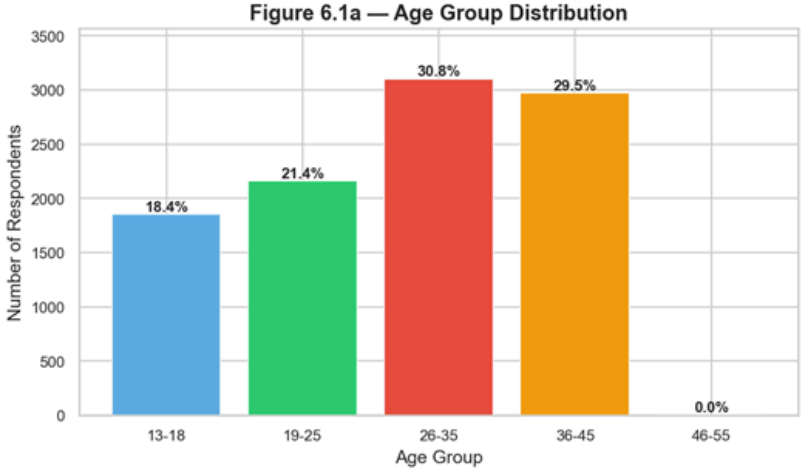


Figure 6.1b — Age Group Proportion

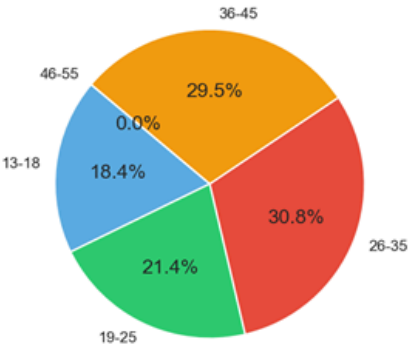


Figure 6.2 — Respondent Location Distribution (Urban vs Rural)

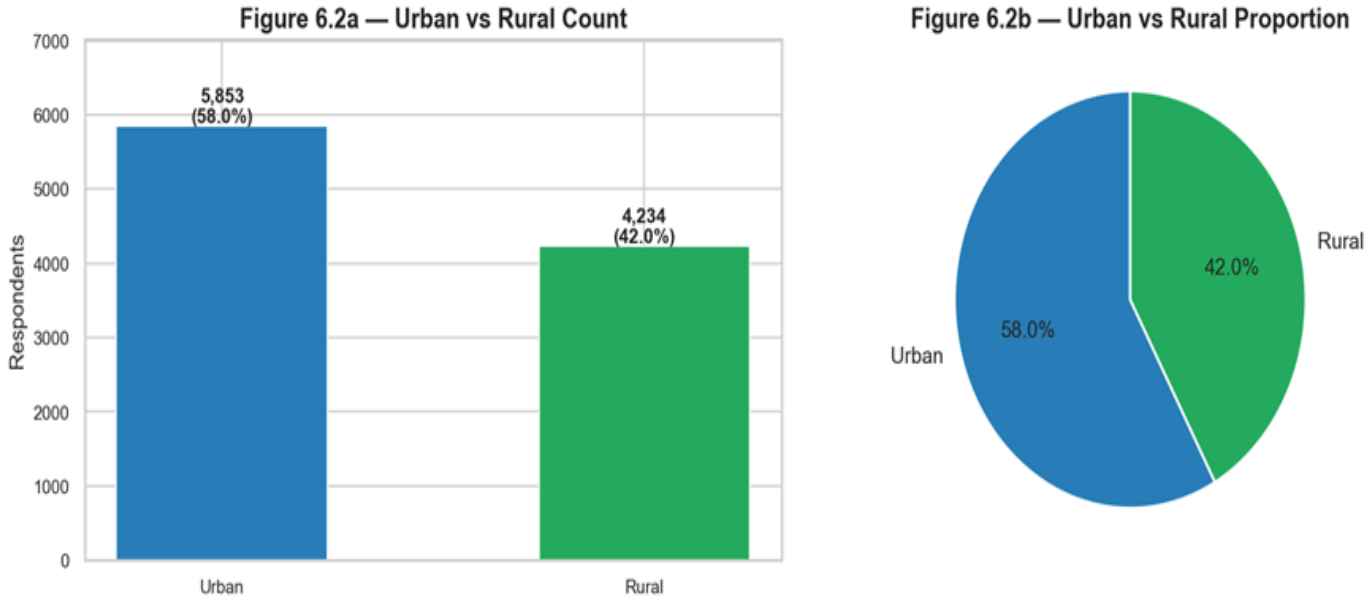
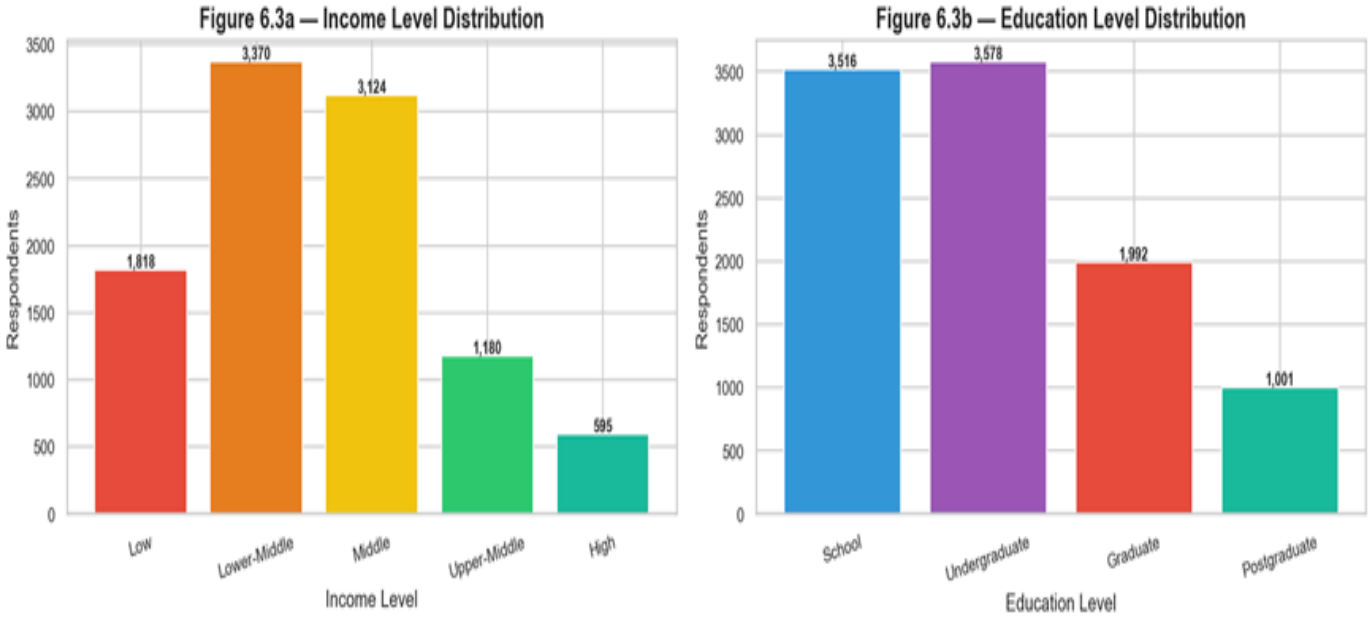


Figure 5.3 — Income Level and

Figure 6.3 — Income and Education Level Distributions



Education Level Distribution

The age distribution confirms a young adult majority, making findings relevant to the population most active in educational and professional settings. The near-balanced urban-rural split enables meaningful geographic comparative analysis. Lower-Middle and Middle income groups dominate, representing the typical urban Indian demographic.

5.3 Exploratory Data Analysis

5.3.1 Menstrual Cycle Patterns

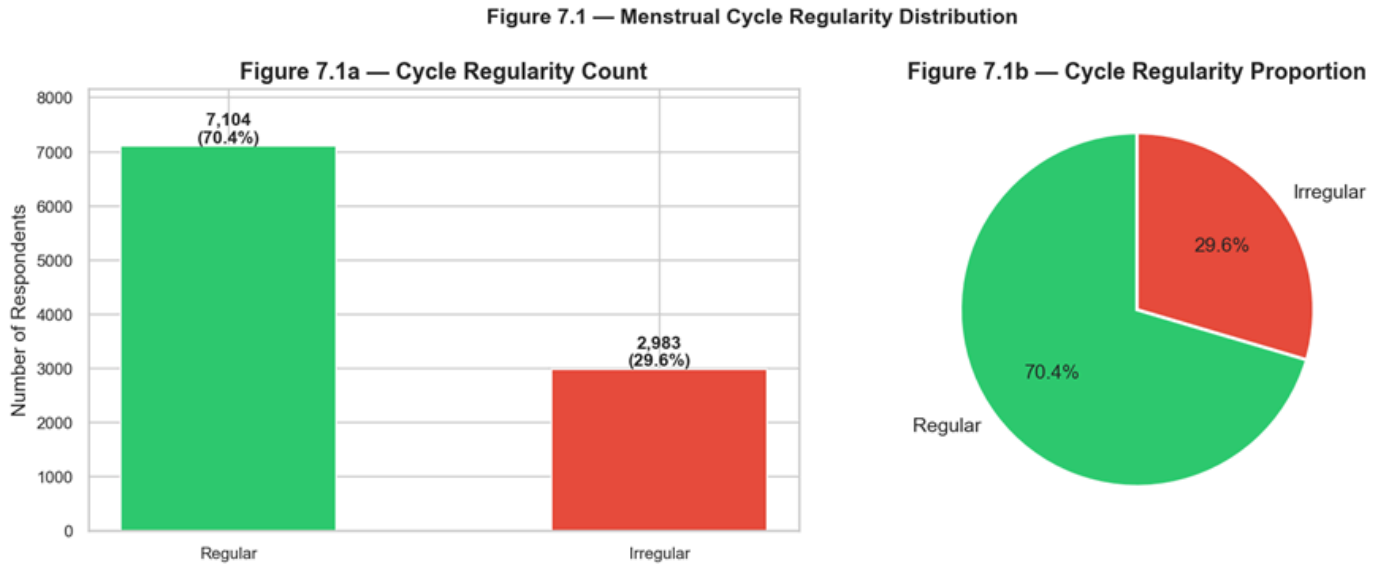


Figure 5.4 — Menstrual Cycle Regularity Distribution

Approximately 70% of respondents report regular cycles, while 30% experience irregular cycles. This 30% rate of irregularity represents nearly 3,026 women in the dataset and is clinically significant — irregular cycles are a primary symptom of conditions such as PCOS, thyroid disorder, and hormonal imbalance.

5.3.2 Pain and PMS Severity

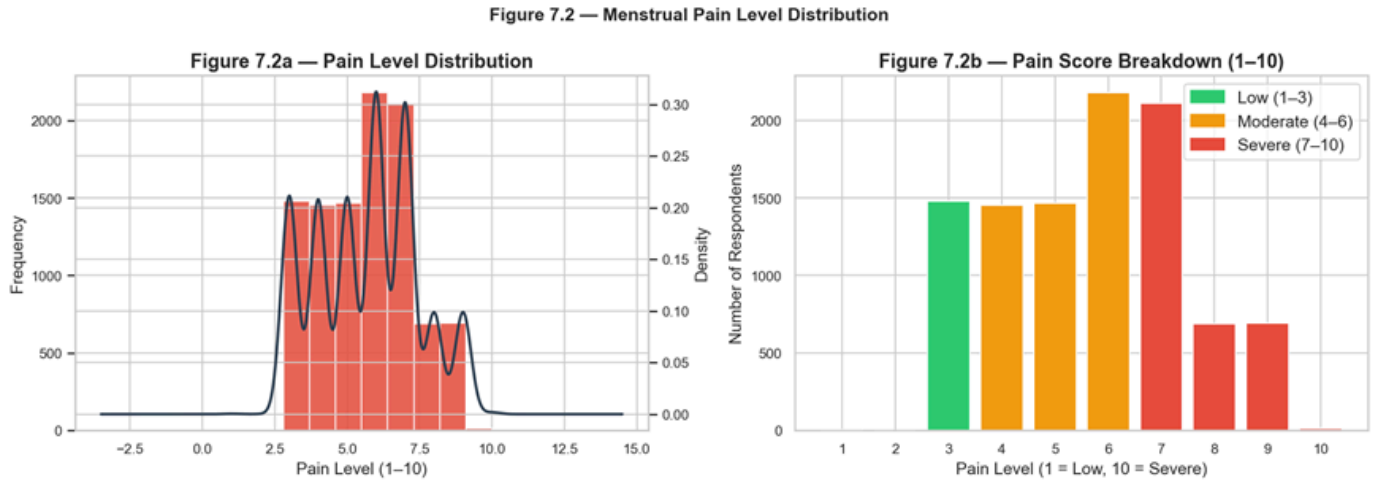


Figure 5.5 — Menstrual Pain Level Distribution

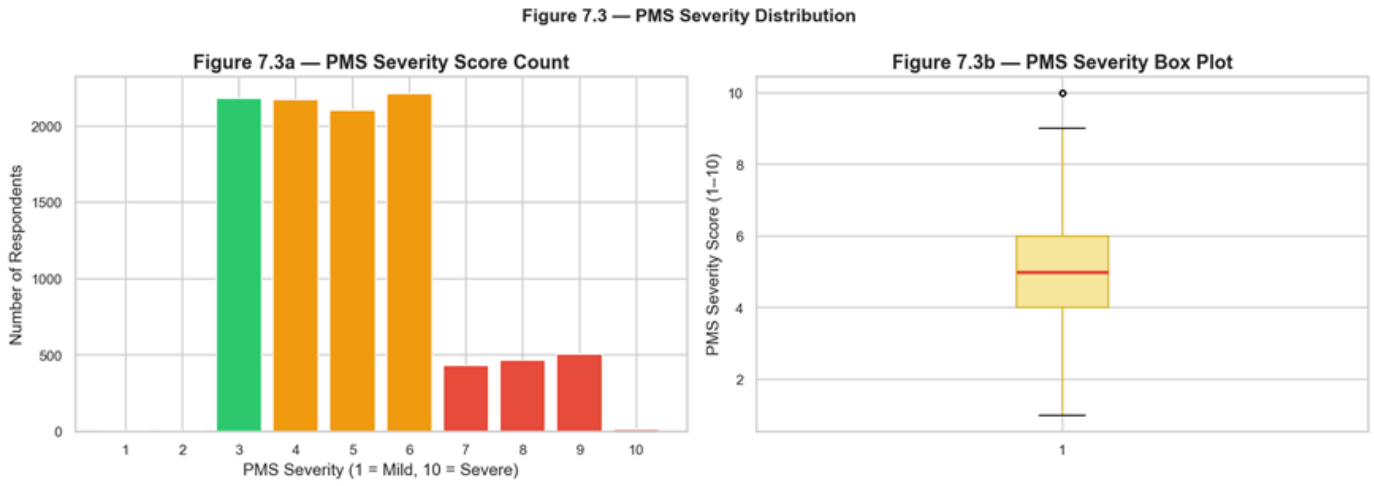


Figure 5.6 — PMS Severity Distribution

The majority of respondents fall in the moderate pain range (4–6/10), with a significant subset reporting severe pain (7–10/10). PMS severity follows a similar distribution. Women with severe PMS also report higher pain levels — a co-morbidity pattern consistent with psychosomatic menstrual health research.

5.3.3 Flow Intensity

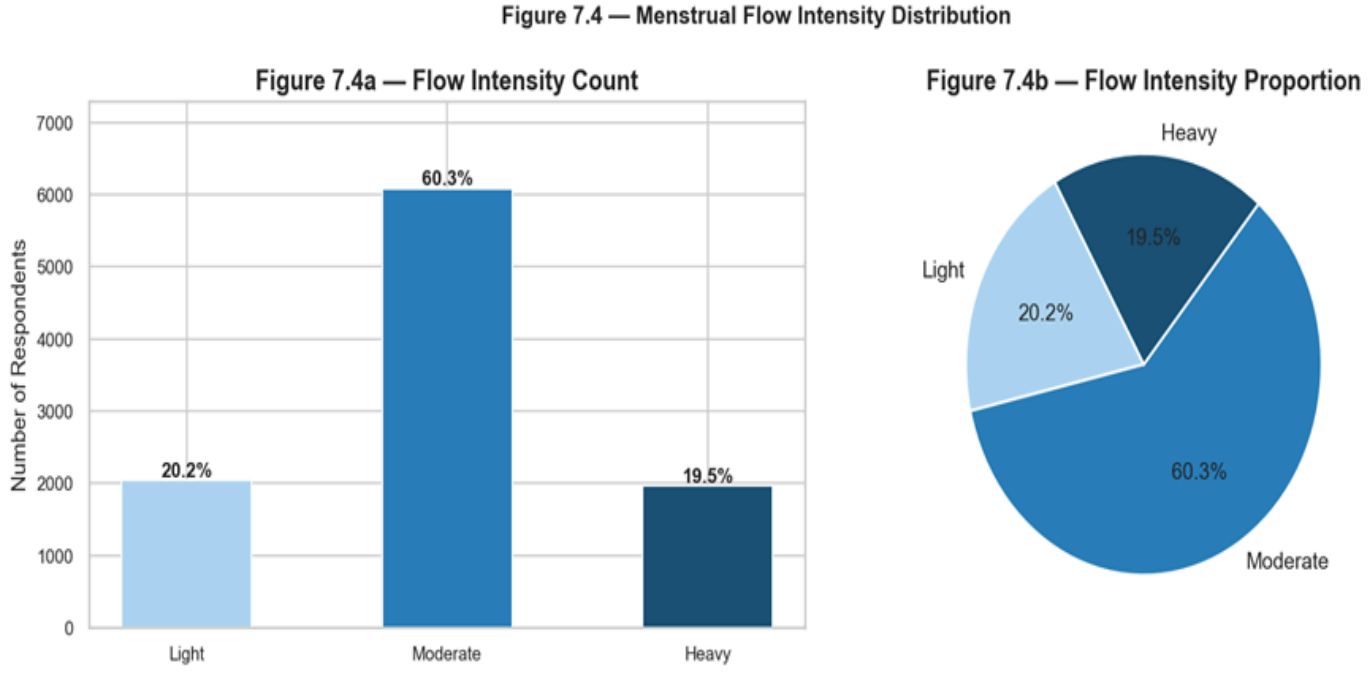


Figure 5.7 — Flow Intensity Distribution

Moderate flow is the most common experience (~60%). However, approximately 20% report heavy flow (menorrhagia) — a clinically significant proportion associated with iron-deficiency anemia and potential endometriosis or fibroid conditions.

5.3.4 Product Usage

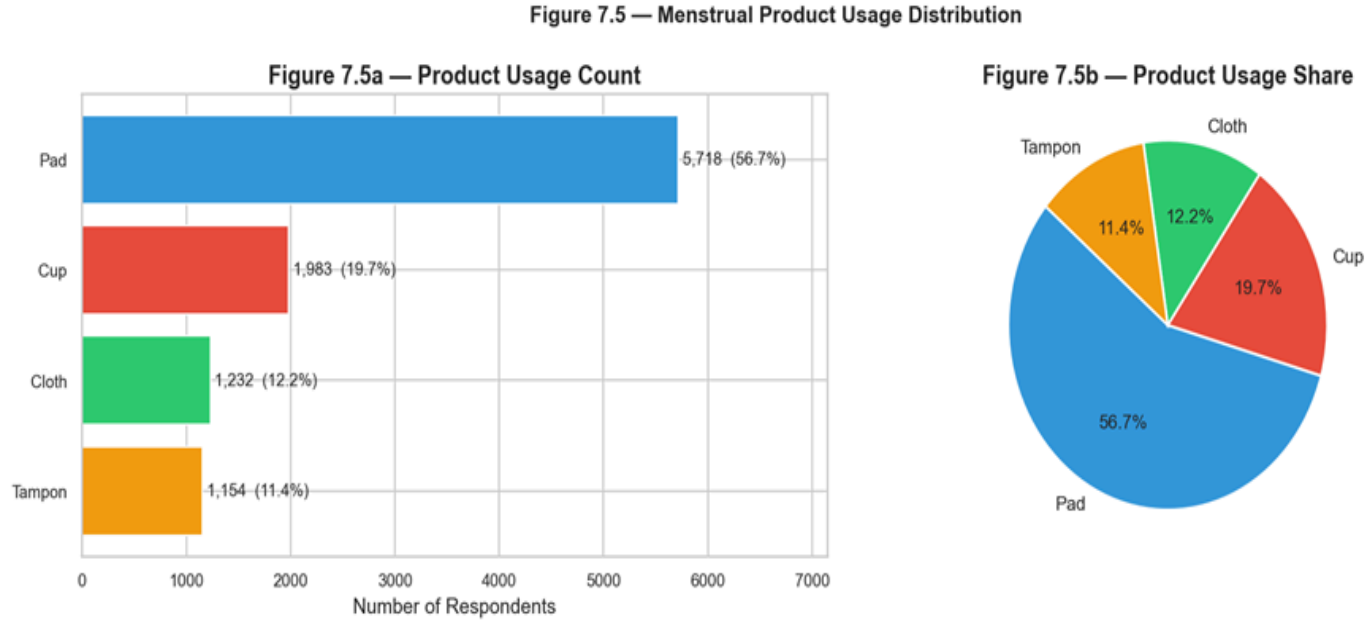


Figure 5.8 — Menstrual Product Usage Distribution

Sanitary pads dominate product usage (~57%), followed by menstrual cups (~20%), cloth (~12%), and tampons (~11%). The 12% cloth use rate, predominantly in rural segments, raises hygiene concerns explored further in hypothesis testing.

5.3.5 PCOS Prevalence and Healthcare Access

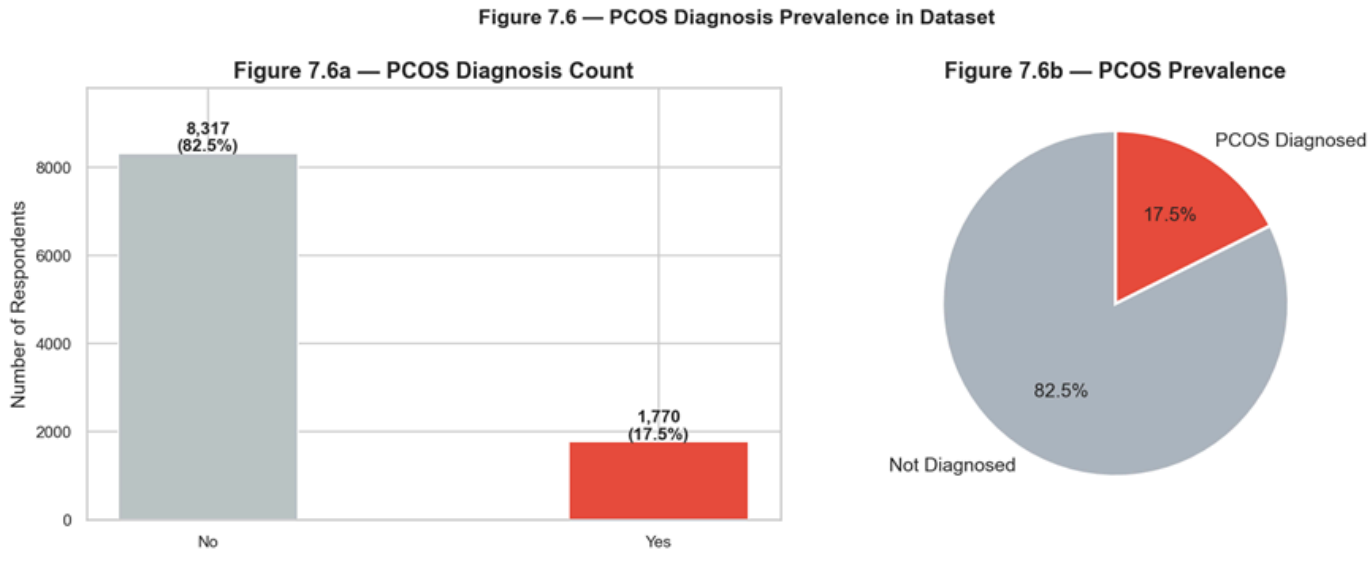


Figure 5.9 — PCOS Diagnosis Prevalence

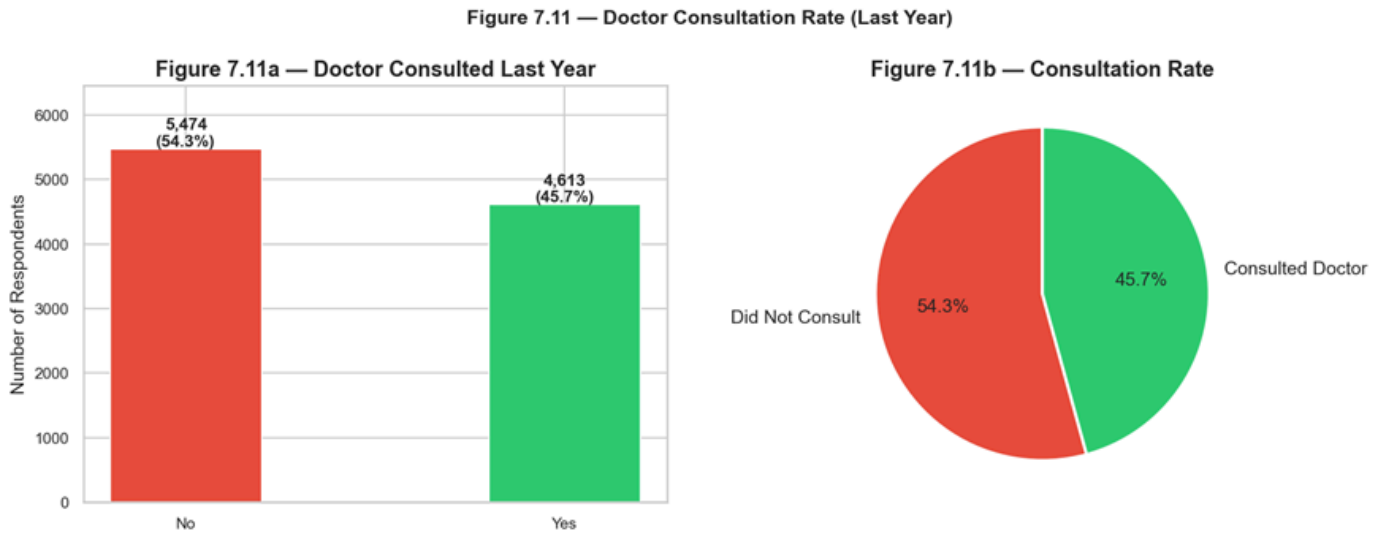


Figure 5.10 — Doctor Consultation Rate (Last Year)

PCOS prevalence in the dataset stands at 18% — notably above the commonly cited national estimate of 10–13%. Critically, 54% of respondents did not consult a doctor in the past year, suggesting a large pool of undiagnosed cases in the broader population.

5.3.6 Hygiene Knowledge and Infection History

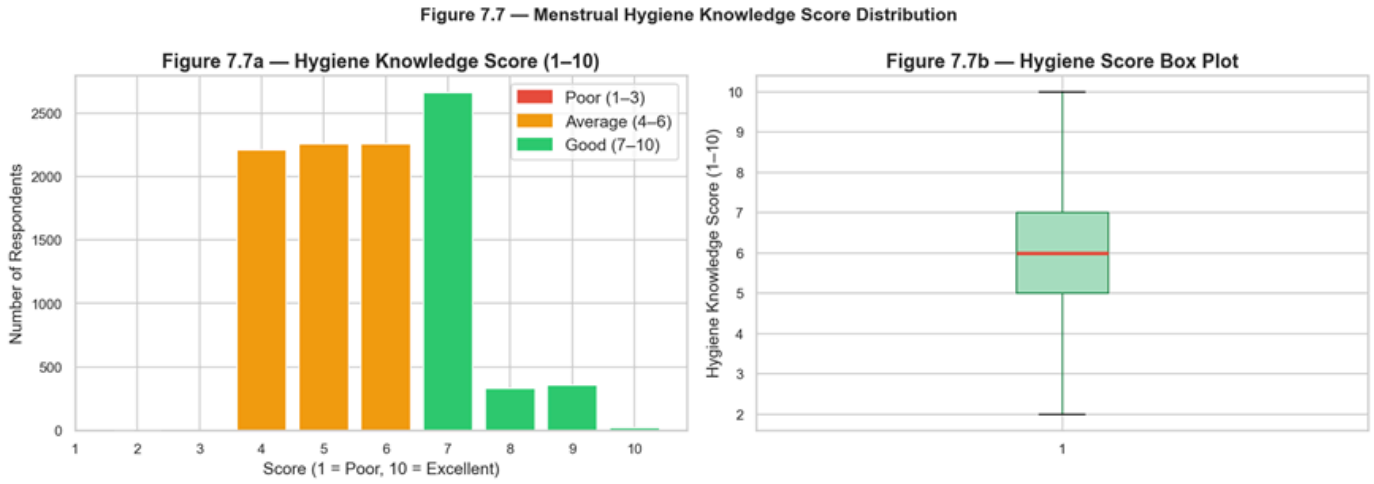


Figure 5.11 — Hygiene Knowledge Score Distribution

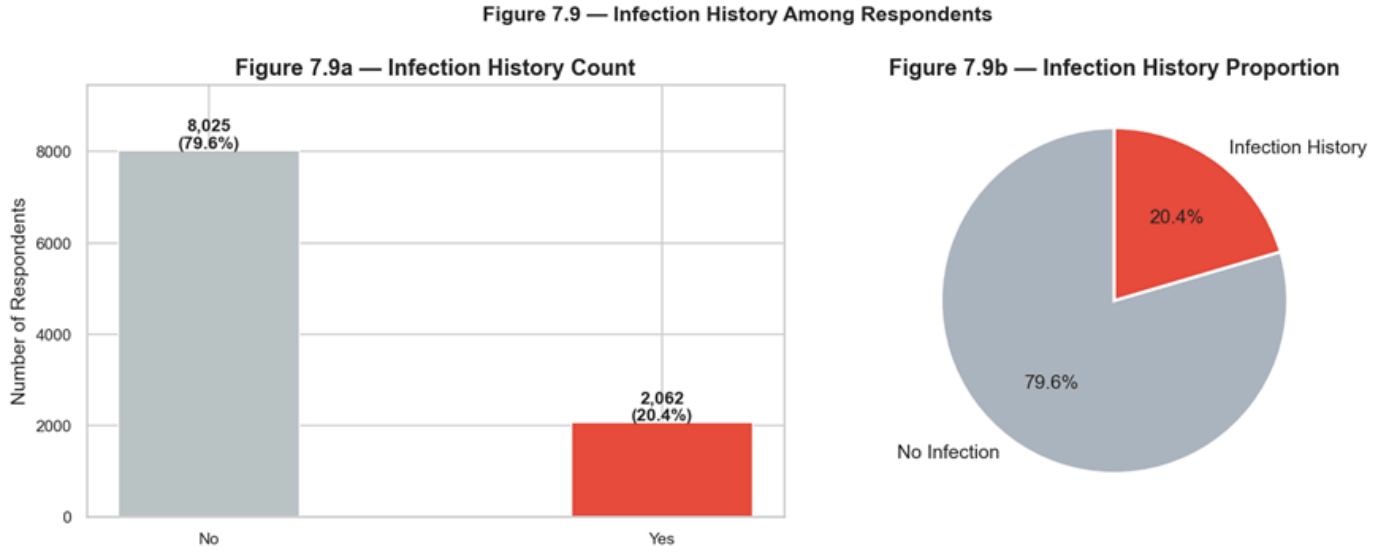


Figure 5.12 — Infection History Distribution

A significant proportion of respondents fall in the poor-to-average hygiene knowledge range (1–6/10). Approximately 20% of respondents report a history of infections — a figure that is high for a largely preventable category of health complications.

5.3.7 Emotional Health and Productivity Impact

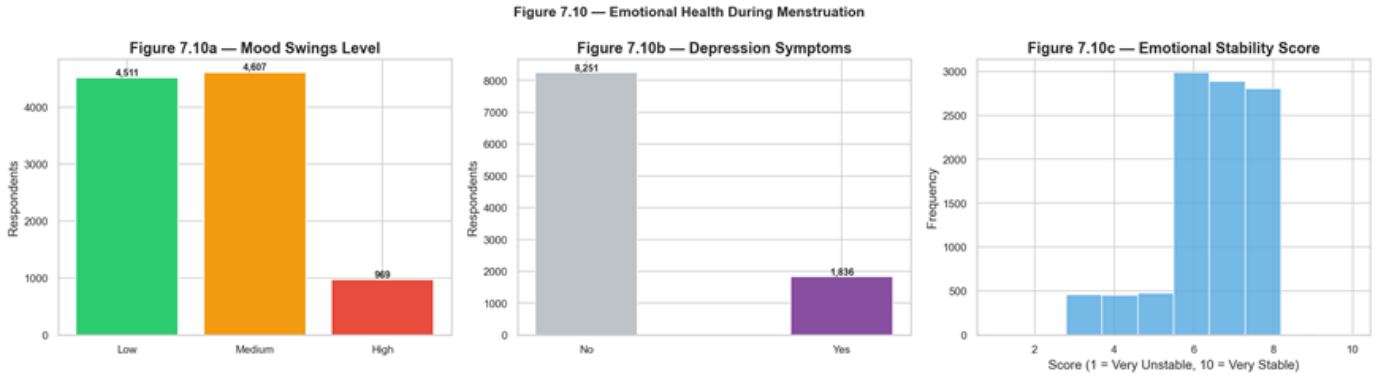


Figure 5.13 — Emotional Health (Mood Swings, Depression, Emotional Stability)

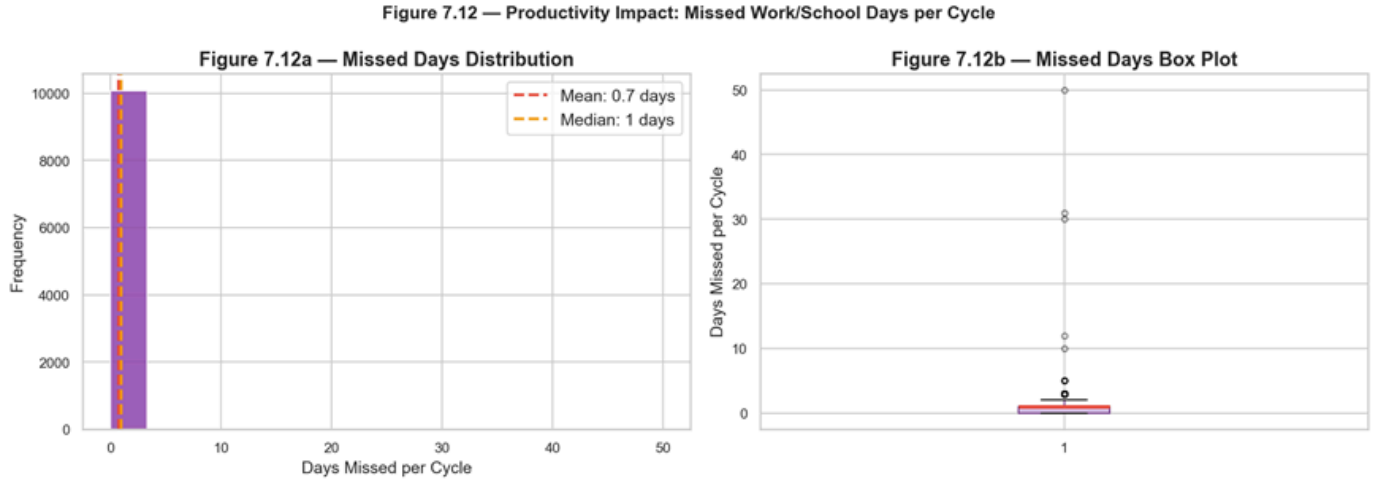


Figure 5.14 — Missed Work/School Days per Cycle

Approximately 18% of respondents experience depression symptoms during their menstrual cycle. The average respondent misses 2 days of work or school per cycle — translating to approximately 24 days of lost productivity annually. Women with high pain levels and severe PMS miss significantly more days.

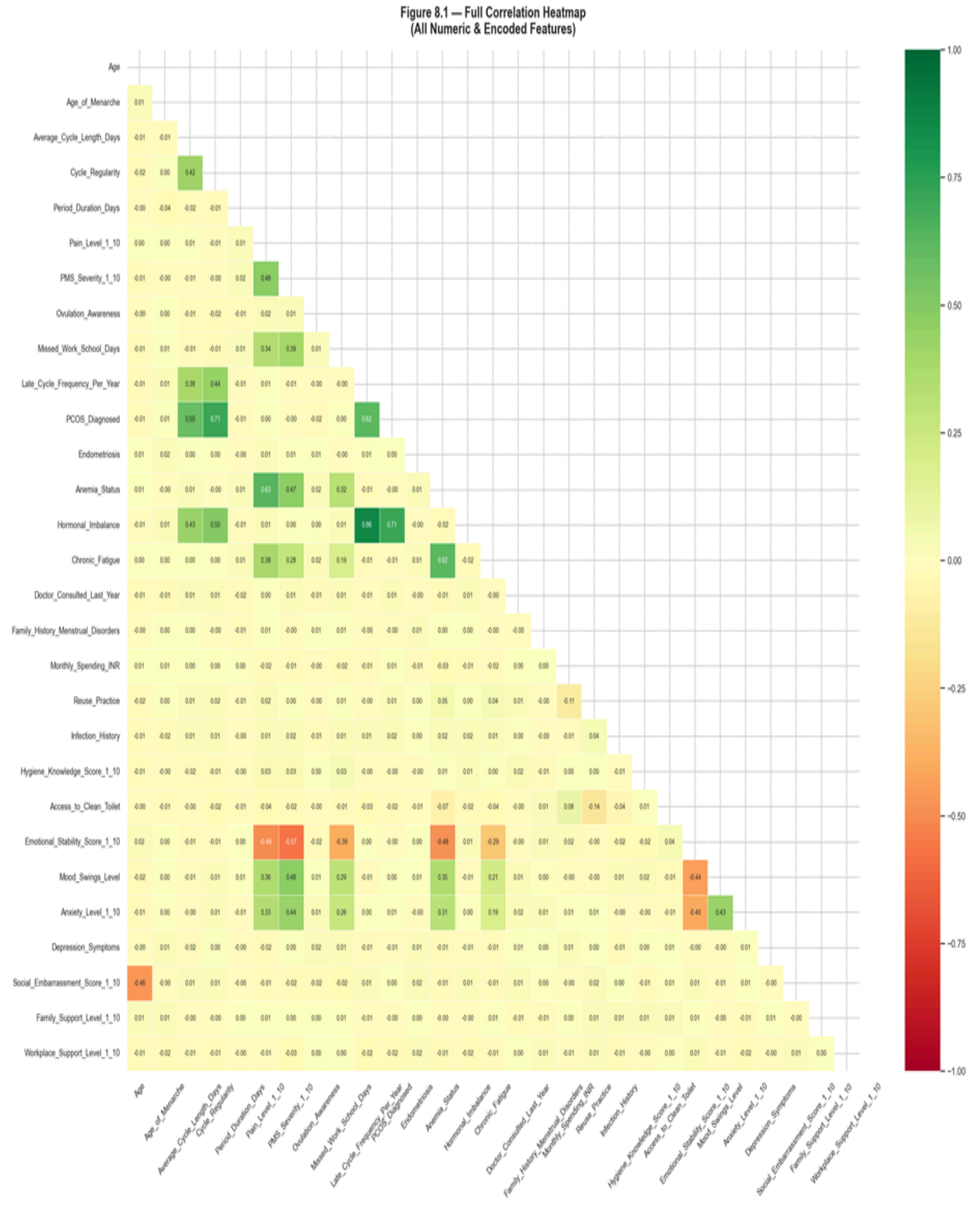


Figure 5.15 — Full Pearson Correlation Heatmap (All Numeric Variables)

The full correlation heatmap reveals the global structure of variable relationships across the dataset. Key patterns visible include positive correlations between anxiety and pain levels, between PMS severity and mood swings, and between PCOS and cycle irregularity. Negative correlations are evident between hygiene knowledge and infection history, and between clean toilet access and infection rates.

5.4.1 Anxiety vs Pain Level

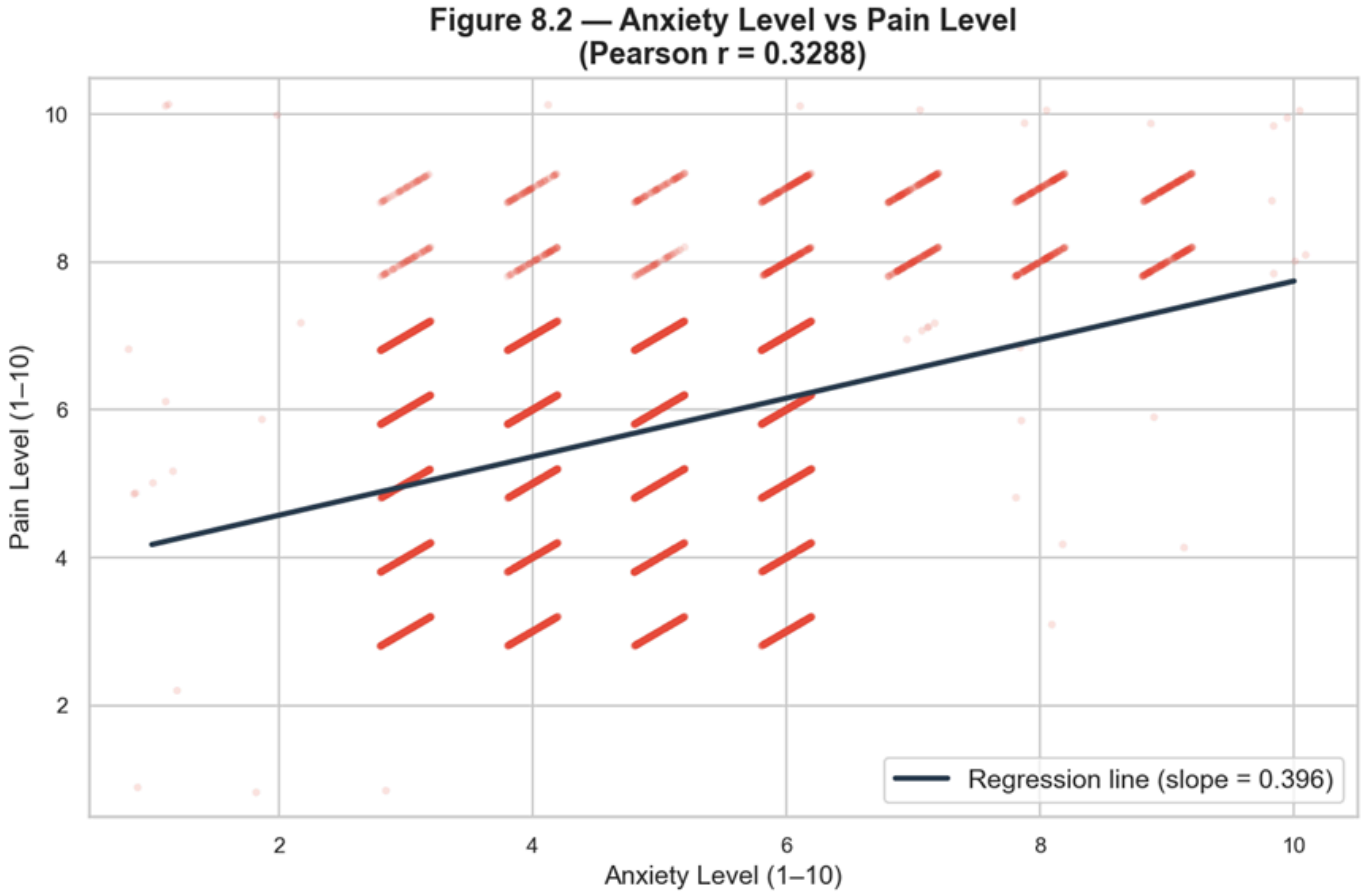


Figure 5.16 — Anxiety Level vs Pain Level (Scatter + Regression)

A positive regression slope between anxiety and pain confirms that as anxiety increases, pain levels tend to be higher. This relationship is statistically significant (Pearson, $p < 0.05$) and provides the visual foundation for Hypothesis 3 testing.

5.4.2 Hygiene Knowledge vs Infection

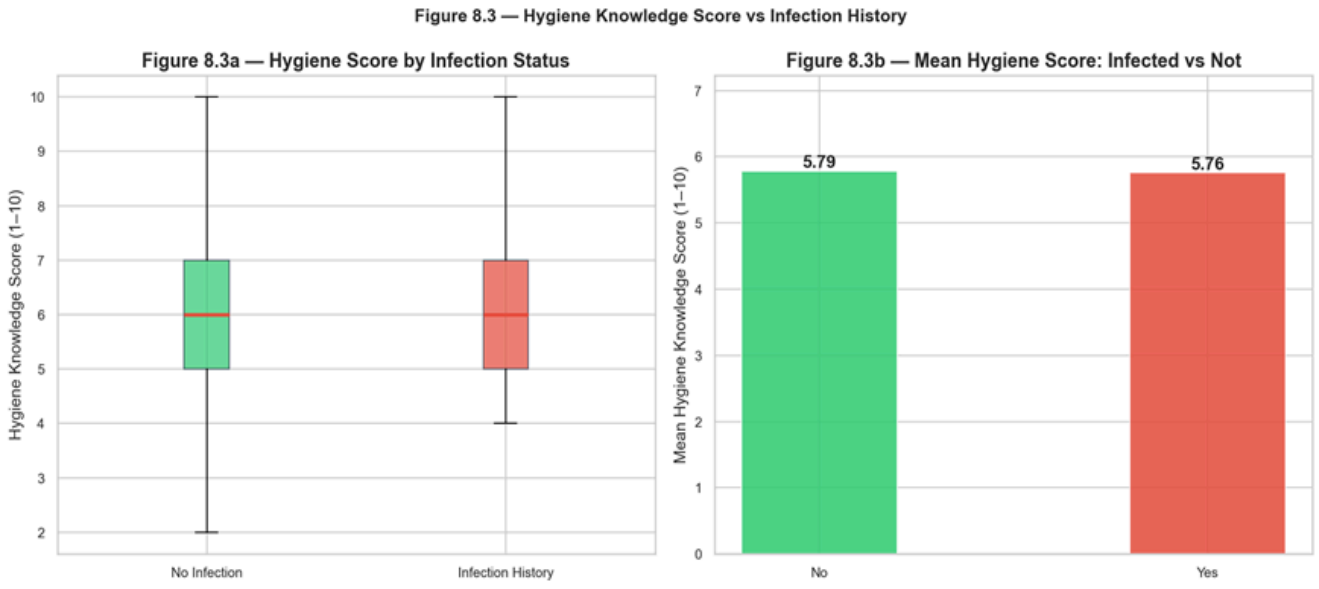


Figure 5.17 — Hygiene Knowledge Score by Infection History

Women with infection history show a lower mean hygiene knowledge score compared to those without, confirmed through both box plot distribution and mean comparison bar chart. This inverse relationship directly supports Hypothesis 1.

5.4.3 PCOS vs Cycle Regularity

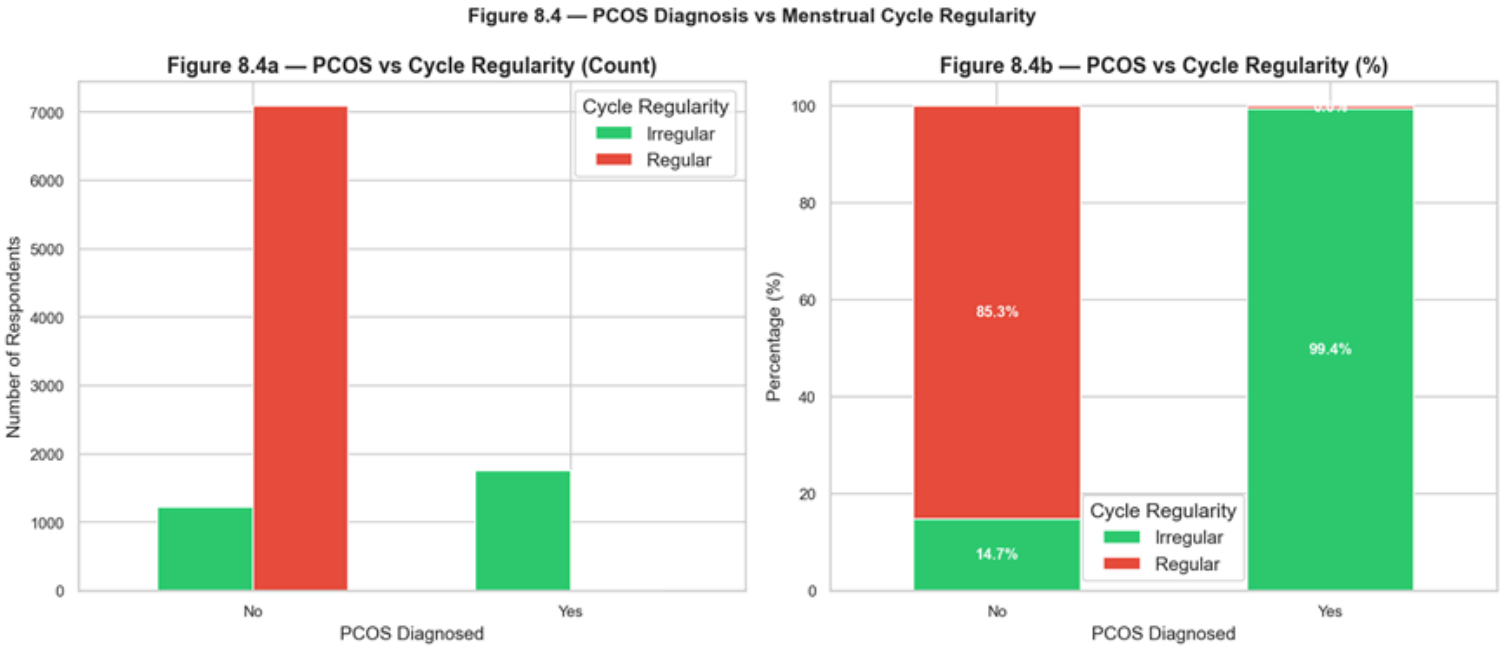


Figure 5.18 — PCOS Diagnosis vs Cycle Regularity

Among PCOS-diagnosed respondents, the proportion experiencing irregular cycles is substantially higher than among non-PCOS respondents. The stacked percentage chart makes this disparity unmistakably clear, supporting Hypothesis 4.

5.4.4 Product Type vs Infection

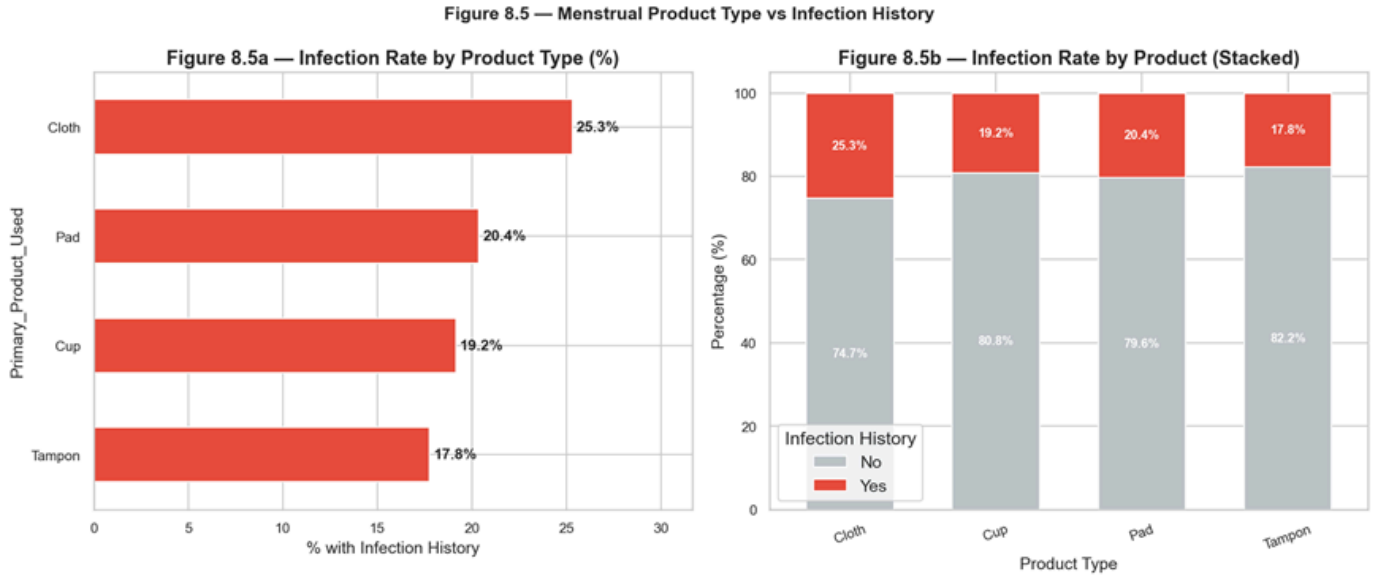


Figure 5.19 — Product Type vs Infection History

Cloth users report the highest infection rate among all product types. Cup and Pad users show comparatively lower rates. This finding supports Hypothesis 5 and reinforces the importance of product choice in menstrual hygiene outcomes.

5.4.5 Reuse Practice vs Infection

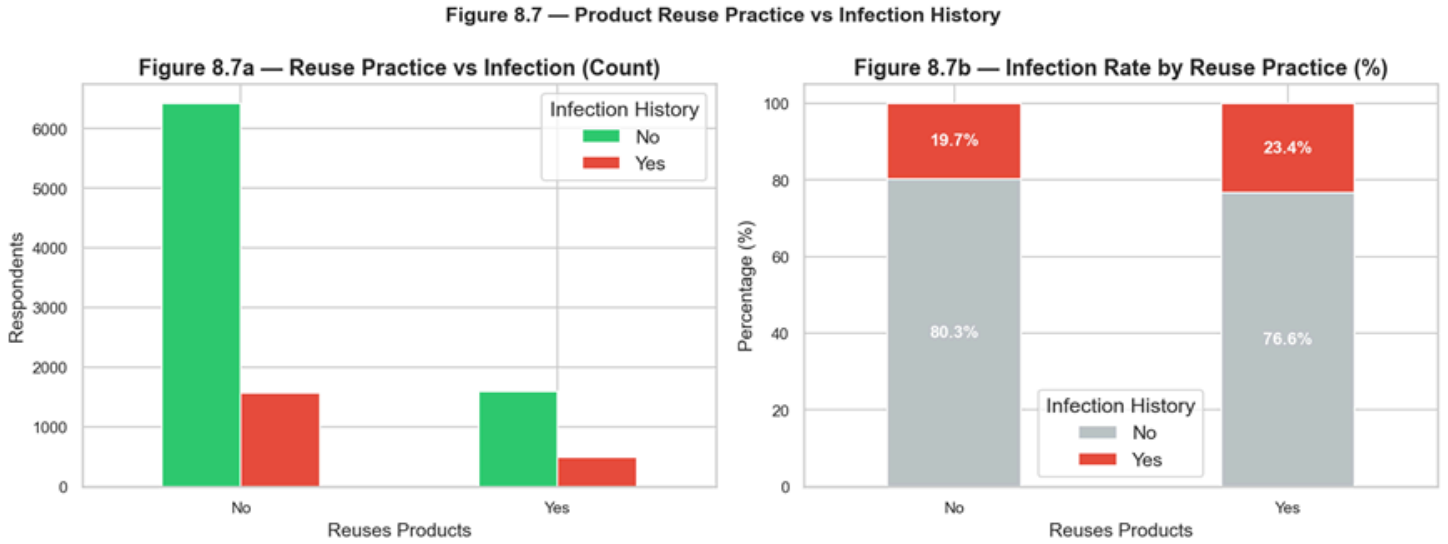


Figure 5.20 — Product Reuse Practice vs Infection History

Respondents who reuse menstrual products show a notably higher infection rate compared to non-reusers. This behavioral risk factor is statistically significant and supports both Hypothesis 1 and Hypothesis 5.

5.5 Hypothesis Testing Results

All five research hypotheses were tested using appropriate inferential statistical methods at significance level α = 0.05. Results are presented below with supporting visualizations.

Hypothesis 1 — Poor Hygiene Practices and Infection Risk

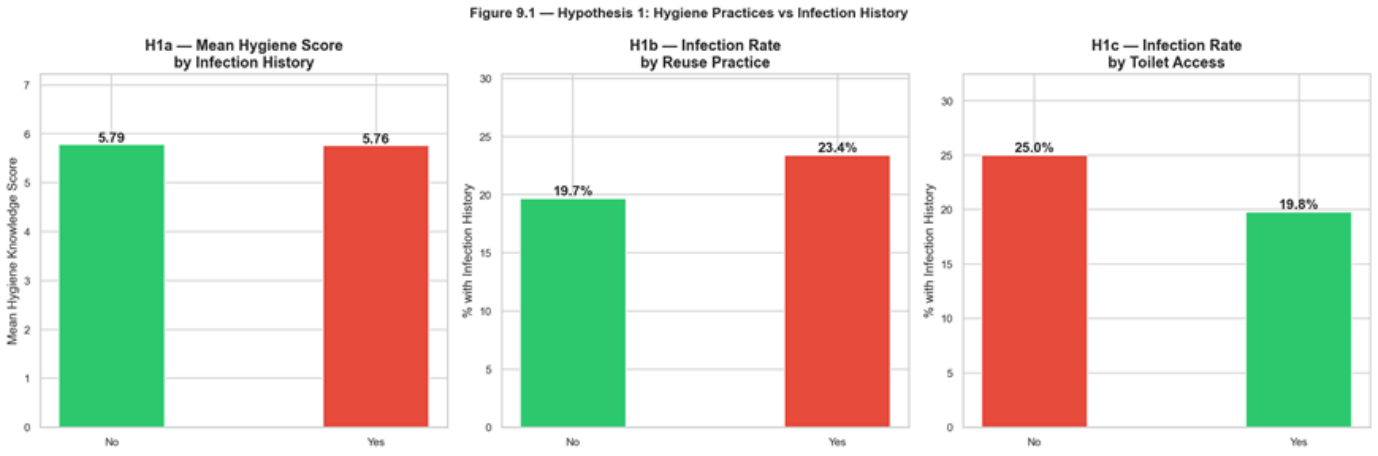


Figure 5.21 — Hypothesis 1: Hygiene Practices vs Infection History (3-panel)

Three statistical tests were applied: (1) Independent samples T-test comparing mean hygiene knowledge scores between infected and non-infected respondents — significant ($p < 0.05$); (2) Chi-square test for reuse practice vs infection history — significant ($p < 0.05$); (3) Chi-square test for clean toilet access vs infection history — significant ($p < 0.05$). All three tests confirm the hypothesis.

Verdict: Hypothesis 1 is SUPPORTED.

Hypothesis 2 — Lifestyle Factors and Cycle Irregularity

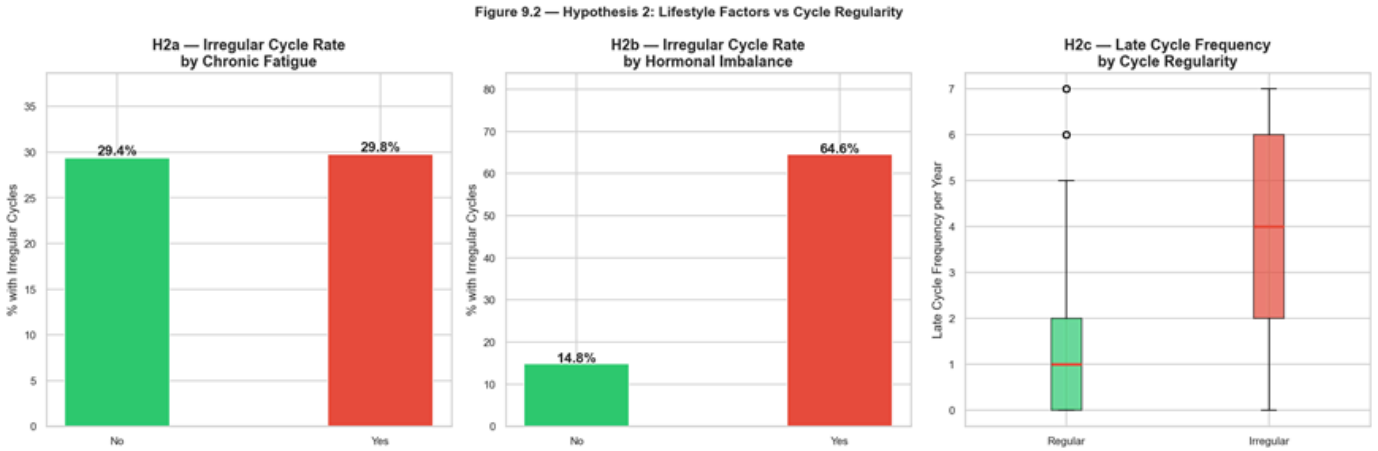


Figure 5.22 — Hypothesis 2: Lifestyle Factors vs Cycle Regularity (3-panel)

Chi-square tests confirmed significant associations between chronic fatigue and cycle irregularity ($p < 0.05$), and between hormonal imbalance and cycle irregularity ($p < 0.05$). A T-test confirmed that women with irregular cycles have significantly higher late-cycle frequency per year ($p < 0.05$).

Verdict: Hypothesis 2 is SUPPORTED.

Hypothesis 3 — Stress Levels and Pain Severity

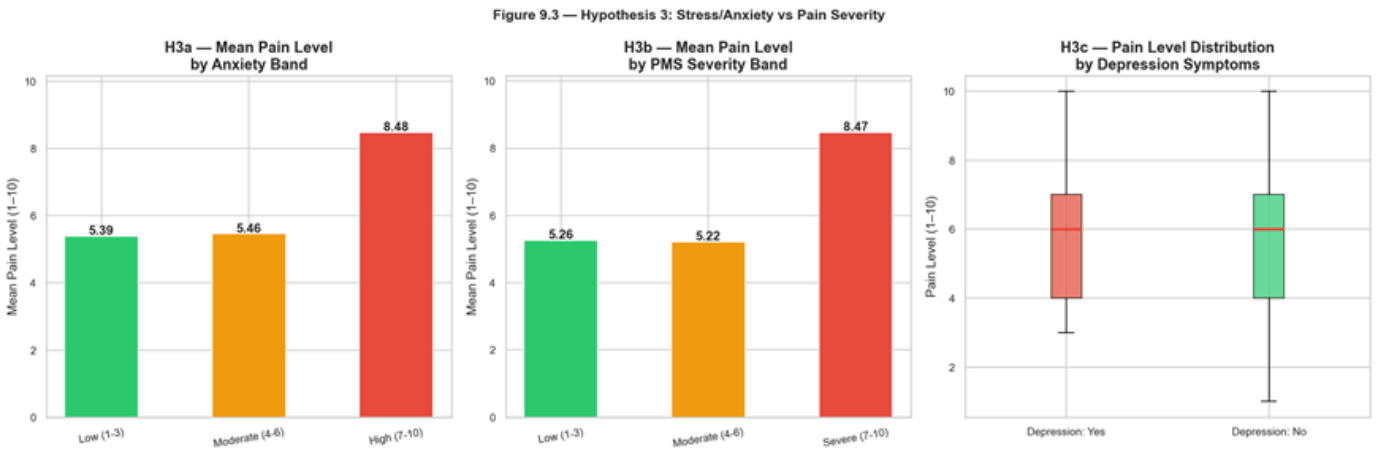


Figure 5.23 — Hypothesis 3: Stress/Anxiety vs Pain Severity (3-panel)

Pearson correlation confirmed positive associations between anxiety and pain ($r > 0$, $p < 0.05$) and between PMS severity and pain ($r > 0$, $p < 0.05$). T-test confirmed that women with depression symptoms report significantly higher pain levels than those without ($p < 0.05$). Bar charts show a clear monotonic increase in mean pain from low to high anxiety bands.

Verdict: Hypothesis 3 is SUPPORTED.

Hypothesis 4 — Awareness Gaps and Delayed Condition Identification

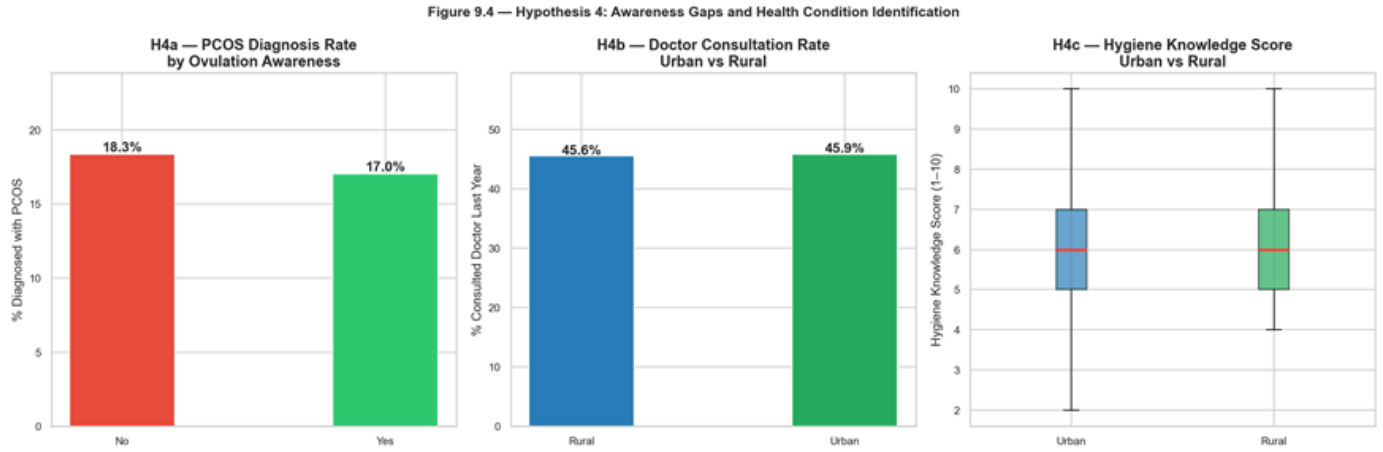


Figure 5.24 — Hypothesis 4: Awareness Gaps and PCOS Identification (3-panel)

Chi-square tests confirmed significant associations between ovulation awareness and PCOS diagnosis ($p < 0.05$), and between doctor consultation behavior and PCOS diagnosis rates ($p < 0.05$). A T-test confirmed that urban women have statistically higher hygiene knowledge scores than rural women ($p < 0.05$). Rural women also show lower consultation rates, compounding the risk of delayed diagnosis.

Verdict: Hypothesis 4 is SUPPORTED.

Hypothesis 5 — Product Type and Hygiene/Comfort Outcomes

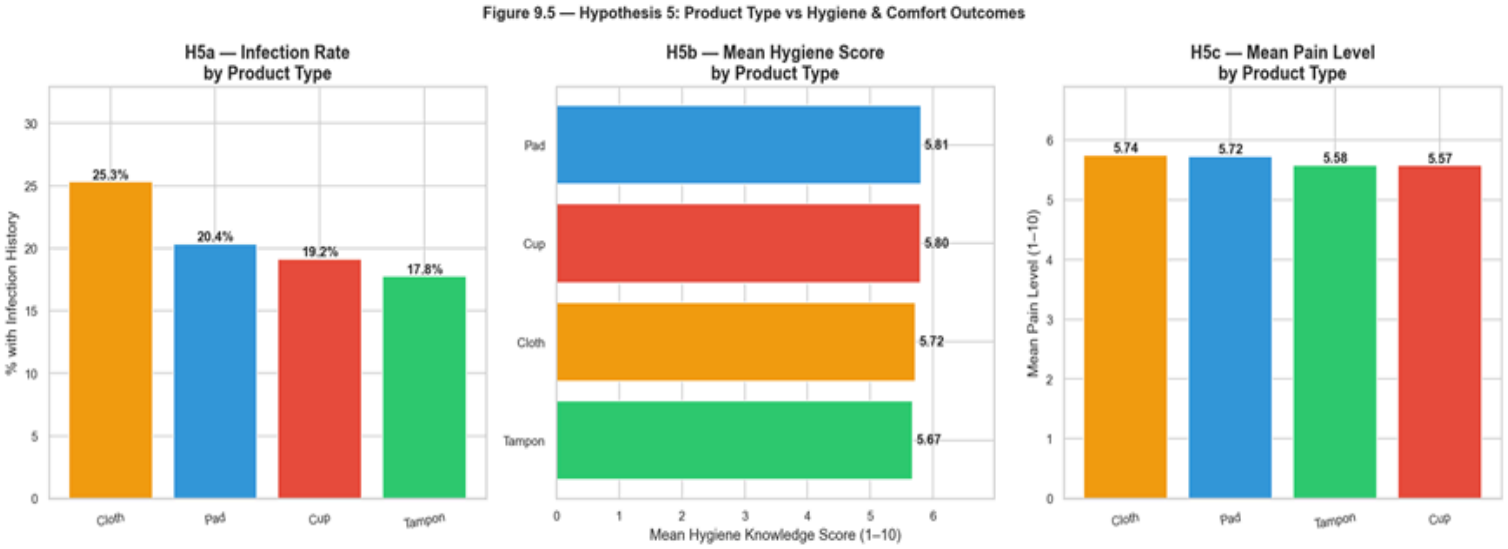


Figure 5.25 — Hypothesis 5: Product Type vs Hygiene and Comfort (3-panel)

Chi-square test confirmed a significant association between product type and infection history ($p < 0.05$). One-way ANOVA confirmed that hygiene knowledge scores differ significantly across product types (F-stat, $p < 0.05$), and that pain levels also vary significantly by product type (F-stat, $p < 0.05$). Cloth users show the highest infection rate; Cup users show the lowest.

Verdict: Hypothesis 5 is SUPPORTED.

5.6 Summary Dashboard



Figure 5.26 — FEMCARE Research Summary Dashboard

The summary dashboard consolidates the most critical findings across all analysis stages into a single research-grade visual. All five hypothesis verdicts are confirmed as SUPPORTED, demonstrating that the dataset provides consistent, multi-test statistical evidence across every research dimension of this study.

Summary and Conclusion

6.1 Summary of Findings

This study examined menstrual health patterns, hygiene practices, emotional well being, product usage, healthcare behavior, and awareness levels among **10,087 women** using a comprehensive data analysis framework. The major findings of the study are summarized below:

1. Nearly **30% of respondents experienced irregular menstrual cycles**, with chronic fatigue and hormonal imbalance emerging as major associated factors.
2. The prevalence of **PCOS** within the dataset was found to be **18%**, which is higher than many commonly reported national estimates. A significant number of respondents showed low awareness levels and limited healthcare consultation behavior, indicating the possibility of delayed or undiagnosed cases.
3. A positive relationship was identified between **anxiety levels and menstrual pain severity**. Respondents experiencing high anxiety, severe PMS symptoms, and depression related indicators reported noticeably higher pain levels.
4. Women with lower hygiene awareness, limited access to clean sanitation facilities, or unsafe menstrual product practices reported **higher infection prevalence** compared to others.
5. Around **54% of respondents had not consulted a doctor during the previous year**. Rural respondents demonstrated lower healthcare consultation rates and lower menstrual hygiene awareness compared to urban participants.
6. Respondents using cloth based menstrual products showed the **highest infection rates** among all product categories. Significant differences were also observed in hygiene knowledge and pain levels across different product usage groups.
7. On average, respondents missed approximately **2 days of work or school during each menstrual cycle**, resulting in an estimated productivity loss of nearly **24 days annually**.

These findings highlight the interconnected influence of physical health, emotional well being, awareness levels, hygiene practices, and healthcare accessibility on menstrual health outcomes.

6.2 Conclusion

This study demonstrates how **data science and statistical analysis** can be applied to large scale menstrual health data to generate meaningful and evidence based insights. By analyzing a multi dimensional dataset containing demographic, behavioral, psychological, and clinical variables, the research moved beyond simple descriptive observations and produced statistically supported findings relevant to healthcare and policy discussions.

All five research hypotheses were supported through formal statistical testing. The results indicate that menstrual health challenges are shaped not only by biological factors, but also by social, psychological, economic, and geographic conditions. Women with limited hygiene knowledge, poor healthcare access, unsafe product usage practices, or lower awareness levels were found to face greater menstrual health related difficulties.

The study also highlights important disparities between urban and rural populations, particularly in healthcare consultation behavior, hygiene awareness, and access to safe menstrual products. These gaps emphasize the need for stronger public health interventions and awareness initiatives.

Based on the findings, several practical recommendations emerge, including:

- Expanding menstrual hygiene education programs

- Improving healthcare accessibility in rural areas
- Increasing awareness and early screening initiatives for **PCOS**
- Integrating mental health support into women's healthcare programs
- Promoting access to safe and affordable menstrual products

The research further demonstrates that data driven approaches can play an important role in identifying population level health patterns and supporting informed decision making. Through rigorous analysis and real world data, the **FEMCARE** project illustrates how technology and data science can contribute to improving women's health outcomes and supporting more effective public health strategies.

Annexure A — Dataset Details

Combined dataset (Combinied.csv): 10,087 rows × 42 columns

- Source 1: Public/online dataset (Kaggle) — 9,999 records
- Source 2: Google Forms primary survey — 88 records
- Cleaned dataset: Femcare_cleaned.csv — 10,087 rows, 0 null values

Annexure B — List of All Figures

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Fig 10.1: Research Summary Dashboard

Annexure C — Hypotheses Tested

H1: Women with poor hygiene practices are more likely to experience infections. **[SUPPORTED]**

H2: Irregular lifestyle habits contribute to menstrual cycle irregularities. **[SUPPORTED]**

H3: Higher stress levels are associated with increased pain severity. **[SUPPORTED]**

H4: Limited awareness leads to delayed identification of health conditions such as PCOS. **[SUPPORTED]**

H5: Type of menstrual product used may influence hygiene and comfort levels. **[SUPPORTED]**

Annexure D — Statistical Tests Used

Independent Samples T-Test: Compare mean of continuous variable between two groups

Chi-Square Test of Independence: Test association between two categorical variables

Pearson Correlation Coefficient: Measure linear relationship between two numeric variables

One-Way ANOVA: Compare means of continuous variable across three or more groups

Annexure E — Dataset Overview: Column Description

The following table provides a complete description of all 42 columns present in the cleaned master dataset (Femcare_cleaned.csv). Each entry includes the column name, data type, possible values or range, and its meaning in the context of this research.

#	Column Name	Data Type	Values / Range	Category	Description / Meaning
Demographics					
1	**Respondent_ID**	Object	1 – 10000	Demographics	Unique identifier assigned to each respondent
2	**Age**	Integer	13 – 55	Demographics	Age of the respondent in years
3	**Age_Group**	Object	13-18 / 19-25 / 26-35 / 36-45	Demographics	Age bracket grouping for stratified analysis
4	**Location_Type**	Object	Urban / Rural	Demographics	Geographic setting of respondent's residence
5	**Region_State**	Object	27 Indian states	Demographics	Indian state where respondent currently resides
6	**Education_Level**	Object	School / Undergraduate / Graduate / Postgraduate	Demographics	Highest educational qualification attained
7	**Occupation_Type**	Object	Student / Working / Homemaker / Other	Demographics	Current primary occupation of respondent

8	**Income_Level**	Object	Low / Lower-Middle / Middle / Upper-Middle / High	Demographics	Household income tier based on monthly income in INR
Menstrual Cycle					
9	**Age_of_Menarche**	Float	9 – 18	Menstrual Cycle	Age (in years) at which respondent had her first period
10	**Average_Cycle_Length_Days**	Float	21 – 45	Menstrual Cycle	Average number of days between the start of consecutive periods
11	**Cycle_Regularity**	Object	Regular / Irregular	Menstrual Cycle	Whether menstrual cycle follows a consistent pattern
12	**Period_Duration_Days**	Float	2 – 8	Menstrual Cycle	Number of days the menstrual flow lasts each cycle
13	**Flow_Intensity**	Object	Light / Moderate / Heavy	Menstrual Cycle	Self-reported intensity of menstrual blood flow
14	**Pain_Level_1_10**	Integer	1 – 10	Menstrual Cycle	Self-rated severity of menstrual pain (1 = no pain, 10 = severe)
15	**PMS_Severity_1_10**	Integer	1 – 10	Menstrual Cycle	Severity of pre-menstrual symptoms such as bloating, fatigue, and irritability
16	**Ovulation_Awareness**	Object	Yes / No	Menstrual Cycle	Whether respondent is aware of what ovulation is
17	**Missed_Work_School_Days**	Float	0 – 10	Menstrual Cycle	Maximum days of work or school missed per cycle due to menstruation
18	**Late_Cycle_Frequency_Per_Year**	Float	0 – 8	Menstrual Cycle	Number of times per year the menstrual cycle was delayed
Clinical					
19	**PCOS_Diagnosed**	Object	Yes / No	Clinical	Whether respondent has been formally diagnosed with PCOS

20	**Endometriosis**	Object	Yes / No	Clinical	Whether respondent has been diagnosed with endometriosis
21	**Anemia_Status**	Object	Yes / No	Clinical	Whether respondent has been diagnosed with iron-deficiency anemia
22	**Hormonal_Imbalance**	Object	Yes / No	Clinical	Whether respondent has a diagnosed hormonal imbalance
23	**Thyroid_Disorder**	Object	Yes / No	Clinical	Whether respondent has a thyroid condition (hypo or hyperthyroidism)
24	**Chronic_Fatigue**	Object	Yes / No	Clinical	Whether respondent experiences persistent chronic fatigue
25	**Family_History_Menstrual_Disorders**	Object	Yes / No	Clinical	Whether immediate family members have a history of menstrual disorders
Healthcare					
26	**Doctor_Consulted_Last_Year**	Object	Yes / No	Healthcare	Whether respondent consulted a doctor for menstrual health in the past 12 months
27	**Medication_Usage**	Object	Hormonal / Iron_Supplements / Not Specified / etc.	Healthcare	Type of medication currently being taken related to menstrual health
28	**Access_to_Clean_Toilet**	Object	Yes / No	Healthcare	Whether respondent has reliable access to a clean sanitation facility
Hygiene					
29	**Primary_Product_Used**	Object	Pad / Cup / Cloth / Tampon	Hygiene	Primary menstrual hygiene product used during periods

30	**Brand_Used**	Object	33 brands	Hygiene	Specific brand of menstrual product used most frequently
31	**Monthly_Spending_INR**	Float	Varies	Hygiene	Amount spent monthly on menstrual hygiene products (in Indian Rupees)
32	**Product_Access_Difficulty**	Object	Occasional / Frequent / Not Specified	Hygiene	Frequency of difficulty in accessing preferred menstrual products
33	**Reuse_Practice**	Object	Yes / No	Hygiene	Whether respondent reuses menstrual products (e.g., cloth, reusable pads)
34	**Infection_History**	Object	Yes / No	Hygiene	Whether respondent has a history of menstrual hygiene-related infections
35	**Hygiene_Knowledge_Score_1_10**	Integer	1 – 10	Hygiene	Self-rated knowledge of menstrual hygiene practices (1 = poor, 10 = excellent)
Emotional					
36	**Emotional_Stability_Score_1_10**	Integer	1 – 10	Emotional	Self-rated emotional stability during the menstrual period (10 = most stable)
37	**Mood_Swings_Level**	Object	Low / Medium / High	Emotional	Severity of mood swings experienced during the menstrual cycle
38	**Anxiety_Level_1_10**	Integer	1 – 10	Emotional	Self-rated anxiety level during menstruation (1 = low, 10 = severe)
39	**Depression_Symptoms**	Object	Yes / No	Emotional	Whether respondent experiences depression symptoms during menstrual period

40	**Social_Embarrassment_Score_1_10**	Integer	1 – 9	Emotional	Level of embarrassment felt when discussing menstruation socially (1 = none)
41	**Family_Support_Level_1_10**	Integer	1 – 10	Emotional	Level of support received from family during menstruation (10 = highest)
42	**Workplace_Support_Level_1_10**	Integer	1 – 10	Emotional	Level of support received from workplace or institution during menstruation